Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The value of p for which the function $f(x)=\begin{cases}\frac{(4^x-1)^3}{\sin\frac{x}{p}\log\left[1+\frac{x^2}{3}\right]}, x\neq 0\\ 12(\log 4)^3, x=0\end{cases}$ may be continuous at $x=0$, is

(A) 1

(B) 2

(C) 3

(D) None of these

Q.2

Let $f''(x)$ be continuous at $x=0$ and $f''(0)=4$. Then value of $\lim_{x\to 0}\frac{2f(x)-3f(2x)+f(4x)}{x^2}$ is

(A) 12

(B) 10

(C) 6

(D) 4

Q.3

Let $f(x)=\frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{x}$. The value which should be assigned to f at $x=0$ so that it is continuous everywhere is

(A) 1

(B) 2

(C) -2

(D) $1/2$

Q.4

If function $f(x)=\frac{2x-\sin^{-1}x}{2x+\tan^{-1}x}$ is continuous at every point of its domain, then $f(0)$ is equal to

(A) $1/3$

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(B) $-1/3$

(C) $2/3$

(D) 2

Q.5

If $f(x) = \begin{cases} \dfrac{(4^x-1)^3}{\sin\left(\frac{x}{a}\right)\log\left(1+\frac{x^2}{3}\right)} & , x \neq 0 \\ (\log 4)^3 & , x = 0 \end{cases}$ is continuous at $x = 0$, then a is equal to

(A) 3

(B) $1/3$

(C) 1

(D) 2

Q.6

$ff(x) = \begin{cases} \frac{\sin 5x}{x^2+2x}, & x \neq 0 \\ k + \frac{1}{2}, & x = 0 \end{cases}$ is continuous at $x = 0$, then k is equal to

(A) 2

(B) $1/2$

(C) 1

(D) -2

Q.7

The function $f : \mathrm{R}\{0\} \to \mathrm{R}$ given by
$f(x) = \frac{1}{x} - \frac{2}{e^{2x}-1}$ can be made continuous at $x = 0$ by defining $f(0)$ as

(A) 0

(B) 1

(C) 2

(D) -1

Q.8

The function $f(x) = [x]\cos\left(\frac{2x-1}{2}\right)\pi$, where [ ] denotes the greatest integer function, is discontinuous at

(A) all x

(B) all integral points

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(C) no x

(D) x which is not an integer

Q.9

ff $f(x)=\frac{\sin\left(e^{x-2}-1\right)}{\log(x-1)}, x \neq 2$ is continuous at $x=2$, then $f(2)$ is equal to

(A) 0

(B) 2

(C) 1

(D) -2

Q.10

$ff(x)=\frac{1-\sin x}{(\pi-2x)^2}$, when $x \neq \frac{\pi}{2}$ and $f\left(\frac{\pi}{2}\right)=\lambda$, then $\mathrm{f}(\mathrm{x})$ will be a continuous function at x $=\frac{\pi}{2}$, when $\lambda=$

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{1}{8}$

(D) none

Q.11

Let $f(x)=\begin{cases}(1+|\sin x|)^{a/|\sin x|} & -\pi/6<x<0 \\ b, & x=0 \\ e^{\tan 2x/\tan 3x}, & 0<x<-\pi/6\end{cases}$ then the value of a and b if f is contiouous at $x=0$, are respectively

(A) $\frac{2}{3}, \frac{3}{2}$

(B) $\frac{2}{3}, e^{2/3}$

(C) $\frac{3}{2}, e^{3/2}$

(D) None of these

Q.12

Let $f(x)$ be defined for all $x>0$ and be continuous.
Let $f(x)$ satisfy $f\left(\frac{x}{y}\right)=f(x)-f(y)$ for all x, y and $f(e)=1$. Then

(A) $f(x)$ is bounded

(B) $f\left(\frac{1}{x}\right) \rightarrow 0$ as $x \rightarrow 0$

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(C) $xf(x) \to 1$ as $x \to 0$

(D) $f(x) = \log x$

Q.13

The function $f(x) = p[x+1] + q[x-1]$, where $[x]$ is the greatest integer function is continuous at $\mathrm{x} = 1$, if

(A) $\mathrm{p} - \mathrm{q} = 0$

(B) $p + q = 0$

(C) $\mathrm{p} = \mathbf{0}$

(D) $\mathbf{q} = \mathbf{0}$

Q.14

$$f(x) = \begin{cases} -2\sin x & , \quad x \leq -\pi/2 \\ a\sin x + b & , \quad -\pi/2 < x < \pi/2 \\ \cos x & , \quad x \geq \pi/2 \end{cases} \text{ isa continuous function, then}$$

(A) $a = 1, b = 1$

(B) $a = -1, b = 1$

(C) $a = 1, b = -1$

(D) $a = b = -1$

Q.15

The value of b for which the function

$$f(x) = \begin{cases} 5x - 4 & , 0 < x \leq 1 \\ 4x^2 + 3bx & , 1 < x < 2 \end{cases}$$ is continuous at every point of its domain, is

(A) -1

(B) 0

(C) $\frac{13}{3}$

(D) 1

Q.16

$$f(x) = \begin{cases} e^x - |x| & , x \neq 0 \\ 1 & , x = 0 \end{cases}$$

(A) $f(x)$ is discontinuous at $x = 0$

(B) $f(x) < 1$ if $x > 0$

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(C) $f(x) < 1$ if $x < 0$

(D) none of these

Q.17

The number of point at which the function $f(x) = \frac{1}{\log |x|}$ is discontinuous at

(A) 1

(B) 2

(C) 3

(D) 4

Q.18

The function $f(x) = x|x|$ is

(A) Continuous at $x = 0$

(B) discontinuous at $x = 0$

(C) Does not exist at $x = 0$

(D) none of these

Q.19

If $f(x)$ is $\frac{10^x - 5^x - 2^x + 1}{x^2}, x \neq 0$ then for f to be contiouous at $x = 0, f(0)$ should be

(A) $\log 10$

(B) $\log 5 . \log 2$

(C) $\log 5/2$

(D) none of these

Q.20

The set of all points where $\frac{x}{1+|x|}$ is continuous is

(A) $x \geq 0$

(B) $x < 0$

(C) $\mathbf{x} \in \mathbf{R}$

(D) none of these

Answers & Solutions

Q.1 Answer:

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None of these

Solution:

For $f(x)$ to be continuous at $x=0$, we should have $\lim_{x\to 0} f(x) = f(0) = 12(\log 4)^3$

$\lim_{x\to 0} f(x) = \lim_{x\to 0} \left(\frac{4^x-1}{x}\right)^3 \times \frac{\left(\frac{x}{p}\right)}{\left(\sin\frac{x}{p}\right)} \cdot \frac{px^2}{\log\left(1+\frac{1}{3}x^2\right)}$

$= (\log 4)^3 \cdot 1 \cdot \mathrm{p} \cdot \lim_{x\to 0}\left(\frac{x^2}{\frac{1}{3}x^2-\frac{1}{18}x^4+\ldots}\right)$

$= 3\mathrm{p}(\log 4)^3$ **Hence** $\mathrm{p}=4$

Q.2 Answer:

12

Solution:

Given $f''(x)$ is continuous at $x=0 = \lim_{x\to 0} f''(x) = f''(0) = 4$

Now, $\lim_{x\to 0} \frac{2f(x)-3f(2x)+f(4x)}{x^2}$ $\left[\frac{0}{0}\right.$ **form**

$= \lim_{\mathrm{x}\to 0} \frac{2\mathrm{f}'(\mathrm{x})-6\mathrm{f}'(2\mathrm{x})+4\mathrm{f}'(4\mathrm{x})}{2\mathrm{x}}$ $\left[\frac{0}{0}\right.$ **form** $]$

$= \lim_{x\to 0} \frac{2f''(x)-12f''(x)+16f''(4x)}{2}$

[Using L' Hospital Rule successively] $= \frac{2\mathrm{f}''(0)-12\mathrm{f}''(0)+16\mathrm{f}''(0)}{2} = 12$

Q.3 Answer:

1

Solution:

$f(0) = \lim_{x\to 0} \frac{\sqrt{1+\sin x}-\sqrt{1-\sin x}}{x}$ $\left(\frac{0}{0} fom\right)$

$= \lim_{x\to 0}\left(\frac{\cos x}{2\sqrt{1+\sin x}} + \frac{\cos x}{2\sqrt{1-\sin x}}\right) = 1$

Q.4 Answer:

$1/3$

Solution:

$f(x) = \lim_{x\to 0}\left(\frac{2x-\sin^{-1}x}{2x+\tan^{-1}x}\right) = f(0), \quad \left(\frac{0}{0} \text{ form }\right)$

Applying L'Hospital's rule,

$f(0) = \lim_{x\to 0} \frac{\left(2-\frac{1}{\sqrt{1-x^2}}\right)}{\left(2+\frac{1}{1+x^2}\right)} = \frac{2-1}{2+1} = \frac{1}{3}$

Q.5 Answer:

$1/3$

Solution:

$\because f(x)$ **is continuous at** $x=0$

$\therefore \lim_{x\to 0} f(x) = f(0) = (\log 4)^3$

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$$\Rightarrow \lim_{x\to 0} \frac{\left(x\log 4+\frac{x^2(\log 4)^2}{2!}+\ldots\right)^3}{\left(\frac{x}{a}-\frac{x^3}{6a^3}+\ldots\right)\left(\frac{x^2}{3}-\frac{x^4}{18}+\ldots\right)} = (\log 4)^3$$

$$\Rightarrow \frac{(\log 4)^3}{(1/3a)} = (\log 4)^3 \Rightarrow a = 1/3$$

Q.6 Answer:

2

Solution:

$f(x)$ **is contiouous at** $x = 0$

$$\Rightarrow \lim_{x\to 0} f(x) = f(0)$$

$$\Rightarrow \lim_{x\to 0} \frac{\sin 5x}{x^2+2x} = k + \frac{1}{2} \Rightarrow \lim_{x\to 0} \frac{5\cos 5x}{2x+2} = k + \frac{1}{2}$$

$$\Rightarrow \frac{5}{2} = k + \frac{1}{2} \Rightarrow k = 2$$

Q.7 Answer:

1

Solution:

$$f(x) = \frac{1}{x} - \frac{2}{e^{2x}-1}$$

$$= \frac{e^{2x}-1-2x}{x(e^{2x}-1)} = \frac{\left(1+2x+\frac{(2x)^2}{2!}+\ldots\right)-1-2x}{x\left(\left(1+2x+\frac{(2x)^2}{2!}+\ldots\right)-1\right)} = \frac{(2x)^2\left(\frac{1}{2!}+\frac{2x}{3!}+\ldots\right)}{x\cdot(2x)\left(1+\frac{2x}{2!}+\ldots\right)}$$

$$\therefore \lim_{x\to 0} f(x) = \frac{2\cdot(1/2)}{1} = 1$$

Hence, to make $f(x)$ **continuous at** $x = 0$,

$f(0)$ **must be** 1.

Q.8 Answer:

no x

Solution:

By, def., $[x] = k, k \le x \le k+1$ **where k is an integer**

$$\therefore f(x) = k\cos\left(\frac{2x-1}{2}\right)\pi, k \le x \le k+1$$

where $k \in Z$

$$\therefore \lim_{x\to k^-} f(x) = \lim_{h\to 0} f(k-h) = \lim_{h\to 0}(k-1)\cos\left(\frac{2(k-h)-1}{2}\right)\pi = (k-1)\cos\left(\frac{2k-1}{2}\right)\pi = 0$$

and $\lim_{x\to k^+} f(x) = \lim_{h\to 0} f(k+h)$

$$\lim_{h\to\infty} k\cos\left(\frac{2(k+h)-1}{2}\right)\pi = k\cos\left(\frac{2k-1}{2}\right)\pi = 0$$

Also, $f(k) = k\cos\left(\frac{2k-1}{2}\right)\pi = 0$

$\therefore f(x)$ **is continuous at all integer points. Between any two integer points,** $f(x)$ **is a cosine function, hence** $f(x)$ **is everywhere continuous.**

Q.9 Answer:

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1

Solution:

$f(2) = \lim_{x\to 2} \frac{\sin\left(e^{x-2}-1\right)}{(e^{x-2}-1)} \times \frac{\left(e^{x-2}-1\right)}{\log(x-1)}$
$= \lim_{x\to 2} \frac{e^{x-2}-1}{\log(x-1)} \left[\frac{0}{0} \text{ form }\right] \quad = \lim_{x\to 2} \left(e^{x-2}\right)(x-1) = 1$

Q.10 Answer:

$\frac{1}{8}$

Solution:

By LMospital's rule,
$\lim_{x\to \frac{\pi}{2}} f(x) = \lim_{x\to \frac{\pi}{2}} \frac{1-\sin x}{(\pi-2x)^2}$
$= \lim_{x\to \frac{\pi}{2}} \frac{-\cos x}{2(\pi-2x)(-2)} = \frac{1}{8} = \lambda$

Q.11 Answer:

$\frac{2}{3}, e^{2/3}$

Solution:

$$f(x) = \begin{cases} (1+|\sin x|)^{a/\sin x|}, & -\frac{\pi}{6} < x < 0 \\ b, & x = 0 \\ e^{\tan 2x/\tan 3x}, & 0 < x < -\frac{\pi}{6} \end{cases}$$

For $\mathrm{f(x)}$ **to be continuous at** $\mathrm{x} = 0$

$\lim_{x\to 0^-} f(x) = f(0) = \lim_{x\to 0^+} f(x)$

$\lim_{x\to 0^-} (1+|\sin x|)^{a|\sin x|}$

$= e^{x\to 0}\left\{|\sin x| \frac{a}{|\sin x|}\right\} = e^a$

Now, $\lim_{x\to 0^+} e^{\tan 2x/\tan 3x}$

$= \lim_{x\to 0^+} e^{\left(\frac{\tan 2x}{2x} \times 2x\right) / \left(\frac{\tan 3x}{3x} \times 3x\right)}$

$= \lim_{x\to 0^+} e^{2/3} = e^{2/3}$

Since, $f(x)$ **is continuous at** $x = 0$.

$\therefore \mathrm{e^a} = \mathrm{e}^{2/3} \Rightarrow \mathrm{a} = \frac{2}{3}$

and $\mathrm{b} = \mathrm{e}^{2/3}$

Q.12 Answer:

$f(x) = \log x$

Solution:

Taking $f(x) = \log x$**, we see that**

$f\left(\frac{x}{y}\right) = f(x) - f(x)$

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Clearly, $f(x)$ is not bounded and $f\left(\frac{1}{x}\right) =$
$-\log x \to \infty$ as $x \to 0$.
Also, $xf(x) = x\log x \to 0$ as $x \to 0$

Q.13 Answer:

$p + q = 0$

Solution:

$f(x) - p[x+1] + q[x-1]$ and $f(1) = p[1+1] + q[0] = 2p$
This functionwill be continuous at $\mathrm{x} = 1$, then $\mathrm{L}\lim_{x\to 1} f(x) = R\lim_{x\to 1} f(x) = f(1)$
$\Rightarrow \lim_{h\to 0} f(1-h) - \lim_{h\to 0} f(1+h) - f(1)$
$\Rightarrow \lim_{h\to 0} p[1-h+1] + q[1-h-1]$
$\Rightarrow \lim_{h\to 0} p[1+h+1] + q[1+h-1] - f(1)$
$\Rightarrow \lim_{h\to 0} p[2-h] + q[-h] - \lim_{h\to 0} p[2+h] + q[h] = f(1)$
$\Rightarrow \lim_{h\to 0} p(1-h) + q(-h-1) - \lim_{h\to 0}[p(1+h) + q(h-1)] - 2p$
$\Rightarrow p - q = 2p \Rightarrow p + q = 0$

Q.14 Answer:

$a = -1, b = 1$

Solution:

$f(x)$ is a continous finnction, so it is also continous at $x = \pi/2$ and $x = -\pi/2$
Now $f(x)$ is continuous at $x = \pi/2$ and $\Rightarrow f(\pi/2) = f(\pi/2 - 0)$
$\Rightarrow \cos \pi/2 = \lim_{h\to 0}$
$\Rightarrow 0 = a + b$
Also $f(x)$ is continuous at $x = -\pi/2 \Rightarrow f(-\pi/2) = f(-\pi/2 + 0)$
$\Rightarrow -2\sin(-\pi/2) = \lim_{h\to 0}\left[a\sin\left(-\frac{\pi}{2} + h\right) + b\right] \Rightarrow 2 = -a + b$
From [1] and [2], $\mathfrak{a} = -1,\ \mathrm{b} = 1$

Q.15 Answer:

-1

Solution:

$F(1) = 1 = F(1-) = F(1+) = 4 + 3b$
$\Rightarrow b = -1$

Q.16 Answer:

$f(x) < 1$ if $x < 0$

Solution:

$\lim_{x\to 0^+}(e^x - x) = 1$
$\lim_{x\to 0^-}(e^x + x) = 1$

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continuous at $x = 0$

Since e^x **is always less than 1 for negative**

x **and** $1 + |x|$ **is greater than 1 for negative** x

$\Rightarrow f(x) < 1$ **if** $x < 0$

Q.17 Answer:

3

Solution:

$f(x)$ **is not defined at** $x = 0, 1, -1$ **and at all other points** $f(x)$ **is continuous.**

$\therefore \lim_{x \to c} f(x) \neq f(c)$, **if** $c = 0, 1, -1$

$\therefore f(x)$ **is discontinuous at 3 points.**

Q.18 Answer:

Continuous at $x = 0$

Solution:

$f(x) = x|x|$

For $x < 0, |x| = -x \Rightarrow f(x) = -x^2$

For $x \geq 0, |x| = x \Rightarrow f(x) = x^2$

$\therefore \text{LHL} = \lim_{x \to 0} -x^2 = 0, \text{RHL} = \lim_{x \to 0} x^2 = 0$

$f(0) = 0$

$\Rightarrow f(x)$ **is continuous at** $x = 0$

Q.19 Answer:

$\log 5 . \log 2$

Solution:

For f **to be continuous at** $x = 0$,

$f(0) = \lim_{x \to 0} f(x)$

$= \lim_{x \to 0} \frac{10^x - 5^x - 2^x + 1}{x^2} = \lim_{x \to 0} \frac{(5^x - 1)(2^x - 1)}{x^2}$

$= \log 5 \cdot \log 2$

Q.20 Answer:

$x \in R$

Solution:

$|x| \geq 0 \Rightarrow 1 + |x| \geq 1 \Rightarrow$ **Denominator** $\neq 0$

$\therefore f(x) = \frac{x}{1+|x|}$ **is continuous everywhere.**

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